

PRILL-EXCEPTIONAL ÉTALE COVERS FROM RAYNAUD BUNDLES

FDMD-233

ABSTRACT. Let C be a smooth projective connected complex curve of genus $g \geq 2$. We prove that C admits a connected finite étale Galois cover $f: X \rightarrow C$ such that

$$h^0(X, \mathcal{O}_X(f^{-1}(p))) \geq 2$$

for every $p \in C$. Starting with the rank- g^g Raynaud bundle of slope one, we choose a determinant-normalising twist that has no global sections but still acquires a section after twisting by any point of C . The resulting degree-zero bundle has finite monodromy. One of its irreducible monodromy summands has the same pointwise non-vanishing property, and the regular representation of its monodromy group then gives the required cover.

1. INTRODUCTION

Let $f: X \rightarrow C$ be a finite morphism of smooth projective connected complex curves. Prill asked whether a fibre divisor $f^{-1}(p)$ can move in a pencil; equivalently, whether

$$h^0(X, \mathcal{O}_X(f^{-1}(p))) > 1$$

for a general point $p \in C$. The question appears in [1, Chapter VI, Exercise D] and, for curves of genus at least three, is recorded as [5, Problem 14]. Partial affirmative results for Prill's original expectation were obtained in [3, 6]. Landesman and Litt then constructed a connected finite étale cover of degree 36 with this property over every curve of genus two [4].

We call a finite cover *Prill-exceptional* if the inequality holds at every point. Our result gives a uniform construction in every genus at least two.

Theorem 1.1. *Let C be a smooth projective connected complex curve of genus $g \geq 2$. There is a connected finite étale Galois cover $f: X \rightarrow C$ such that*

$$h^0(X, \mathcal{O}_X(f^{-1}(p))) \geq 2$$

for every $p \in C$.

For $g \geq 3$, Theorem 1.1 answers [5, Problem 14] affirmatively. For a finite flat morphism, the function

$$p \mapsto h^0(X, \mathcal{O}_X(f^{-1}(p)))$$

is upper semicontinuous. Hence the locus on which this dimension is at least two is closed; if it contains a dense open subset, it is all of C . Thus the pointwise formulation above is equivalent to the usual general-fibre formulation, while being the form naturally produced by the construction.

The proof uses two properties of Raynaud's bundles. First, the relevant Raynaud bundle has a section after tensoring by any degree-zero line bundle. Second, its pullback under an isogeny of the Jacobian is a direct sum of copies of one line bundle. A carefully chosen twist makes the determinant trivial. On a suitable étale pullback of C , the repeated line bundle is then torsion, which upgrades projectively finite monodromy to finite linear monodromy.

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2. THE RAYNAUD BUNDLE

Write $J = \text{Jac}(C) = \text{Pic}^0(C)$, fix a theta divisor Θ on J , and choose an Abel–Jacobi embedding $i: C \hookrightarrow J$. We use the following standard form of Raynaud’s construction; see [7, Section 3] and [2, Section 2.3].

Theorem 2.1 (Raynaud). *For every integer $n \geq 1$, there is a vector bundle E_n on J such that*

$$[n]^* E_n \cong H^0(J, \mathcal{O}_J(n\Theta))^* \otimes_{\mathbb{C}} \mathcal{O}_J(n\Theta). \quad (1)$$

Its restriction $R_n = i^ E_n$ is semistable, has rank n^g and slope g/n , and satisfies*

$$H^0(C, R_n \otimes \alpha) \neq 0 \quad \text{for every } \alpha \in \text{Pic}^0(C). \quad (2)$$

Take $n = g$ and set

$$R := R_g, \quad r := \text{rk}(R) = g^g.$$

Then R is semistable of slope one, so $\deg(R) = r$.

3. A DETERMINANT-NORMALISED TWIST

The following counting argument produces the twist needed later.

Lemma 3.1. *There is a line bundle $M \in \text{Pic}^{-1}(C)$ such that*

$$M^{\otimes r} \cong (\det R)^{-1} \quad \text{and} \quad H^0(C, R \otimes M) = 0. \quad (3)$$

Proof. The map

$$\text{Pic}^{-1}(C) \longrightarrow \text{Pic}^{-r}(C), \quad M \longmapsto M^{\otimes r},$$

is a translate of the multiplication-by- r isogeny of J . Consequently, the fibre

$$T := \{M \in \text{Pic}^{-1}(C) : M^{\otimes r} \cong (\det R)^{-1}\}$$

has cardinality r^{2g} .

Call $M \in \text{Pic}^{-1}(C)$ *bad* if $H^0(C, R \otimes M) \neq 0$. A nonzero section gives a nonzero morphism $M^{-1} \rightarrow R$. If $L \subset R$ is the saturation of its image, then $L \cong M^{-1}(D)$ for an effective divisor D . Since $\deg(M^{-1}) = 1$ and R is semistable of slope one,

$$1 + \deg D = \deg L \leq 1.$$

Thus $D = 0$, and M^{-1} is a degree-one line subbundle of R .

Fix a Jordan–Hölder filtration $0 = R_0 \subset R_1 \subset \cdots \subset R_m = R$. If $L \subset R$ is a degree-one line subbundle, let k be minimal with $L \subset R_k$. The composite $L \rightarrow R_k/R_{k-1}$ is nonzero. Both bundles are stable of slope one, so this morphism is an isomorphism. Hence L is isomorphic to a rank-one Jordan–Hölder factor of R . There are at most r such factors, counted with multiplicity, and therefore at most r isomorphism classes of bad M .

Because $g \geq 2$ and $r = g^g > 1$, one has $r^{2g} > r$. The set T therefore contains an M that is not bad. $\square$

Fix M as in Lemma 3.1 and put $V := R \otimes M$. Then

$$\deg V = 0, \quad \det V \cong \mathcal{O}_C, \quad H^0(C, V) = 0. \quad (4)$$

For every $p \in C$, the line bundle $M(p)$ has degree zero, so (2) gives

$$H^0(C, V(p)) = H^0(C, R \otimes M(p)) \neq 0. \quad (5)$$

4. FINITE MONODROMY

Proposition 4.1. *The vector bundle V is trivialised by a connected finite étale cover of C .*

Proof. Form the cartesian square

$$\begin{array}{ccc} \tilde{C} & \xrightarrow{\tilde{i}} & J \\ q \downarrow & & \downarrow [g] \\ C & \xrightarrow{i} & J. \end{array}$$

The morphism q is finite étale and surjective. By (1),

$$q^*V \cong H^0(J, \mathcal{O}_J(g\Theta))^* \otimes_{\mathbb{C}} (\tilde{i}^* \mathcal{O}_J(g\Theta) \otimes q^*M).$$

Let $\tilde{C}_0$ be a connected component of $\tilde{C}$. Its image under q is nonempty, open, and closed, hence equals C . Define

$$N := (\tilde{i}^* \mathcal{O}_J(g\Theta) \otimes q^*M)|_{\tilde{C}_0}.$$

Since $h^0(J, \mathcal{O}_J(g\Theta)) = g^g = r$, the restriction of q^*V to $\tilde{C}_0$ is $N^{\oplus r}$. Taking determinants and using $\det V \cong \mathcal{O}_C$ yields

$$N^{\otimes r} \cong \mathcal{O}_{\tilde{C}_0}.$$

Let d be the order of N and choose an isomorphism $N^{\otimes d} \cong \mathcal{O}_{\tilde{C}_0}$. The resulting Kummer cover

$$\mathrm{Spec}_{\tilde{C}_0} \left(\bigoplus_{a=0}^{d-1} N^{-a} \right) \rightarrow \tilde{C}_0$$

is a finite étale μ_d -torsor, and the pullback of N to it is trivial. Each connected component maps surjectively to $\tilde{C}_0$. Composing one such component with $\tilde{C}_0 \rightarrow C$ gives a connected finite étale morphism $h: C' \rightarrow C$ for which h^*V is trivial. $\square$

After replacing this cover by a connected Galois closure, V is the vector bundle associated with a finite-dimensional complex representation of a finite group.

Proposition 4.2. *There is a nontrivial irreducible finite-monodromy vector bundle W on C such that*

$$H^0(C, W(p)) \neq 0 \quad \text{for every } p \in C.$$

Proof. Choose a connected finite étale Galois cover $u: Y \rightarrow C$ trivialising V , and write H for its Galois group. Relative to a trivialisation $u^*V \cong \mathcal{O}_Y^{\oplus r}$, the descent datum is given by matrices whose entries lie in $H^0(Y, \mathcal{O}_Y) = \mathbb{C}$. It is therefore a representation of H . Maschke's theorem gives a decomposition

$$V \cong \bigoplus_{j=1}^s W_j^{\oplus m_j},$$

where the W_j are pairwise nonisomorphic irreducible finite-monodromy bundles. None is trivial, since a trivial summand would contradict $H^0(C, V) = 0$ in (4).

For $1 \leq j \leq s$, set

$$Z_j := \{p \in C : H^0(C, W_j(p)) \neq 0\}.$$

Let $p_1, p_2: C \times C \rightarrow C$ be the projections and let $\Delta \subset C \times C$ be the diagonal. The restriction of

$$p_1^*W_j \otimes \mathcal{O}_{C \times C}(\Delta)$$

to $C \times \{p\}$ is $W_j(p)$. Properness of p_2 and upper semicontinuity therefore show that Z_j is closed. Equation (5) implies $C = \bigcup_{j=1}^s Z_j$. Since C is irreducible and the union is finite, one of the Z_j equals C . $\square$

5. THE COVERING

Proof of Theorem 1.1. Let W be as in Proposition 4.2, and let

$$\rho: \pi_1(C, c) \longrightarrow \mathrm{GL}(W_c)$$

be its monodromy representation. Its image G is finite. The subgroup $\ker \rho$ determines a connected finite étale Galois cover $f: X \rightarrow C$ with Galois group G .

The vector bundle $f_*\mathcal{O}_X$ is associated with the regular representation $\mathbb{C}[G]$. Every irreducible complex representation of G occurs in $\mathbb{C}[G]$, with multiplicity equal to its dimension. In particular, $\mathcal{O}_C$ and W occur as distinct direct summands of $f_*\mathcal{O}_X$.

For $p \in C$, the fibre divisor is the pullback of p , so

$$\mathcal{O}_X(f^{-1}(p)) \cong f^*\mathcal{O}_C(p).$$

The projection formula and the preceding direct-sum decomposition give

$$\begin{aligned} H^0(X, \mathcal{O}_X(f^{-1}(p))) &\cong H^0(C, f_*\mathcal{O}_X \otimes \mathcal{O}_C(p)) \\ &\supset H^0(C, \mathcal{O}_C(p)) \oplus H^0(C, W(p)). \end{aligned}$$

The first summand is nonzero because p is effective; the second is nonzero by Proposition 4.2. Hence the left-hand side has dimension at least two for every $p \in C$. $\square$

Remark 5.1. The construction is not degree-effective. The first bundle already has rank g^g , and the proof subsequently passes to an étale Kummer cover and a Galois closure. The degree-36 construction of [4] in genus two is therefore much sharper numerically.

Remark 5.2. The determinant-normalising step is essential. Formula (1) only makes the pullback of R_g a direct sum of copies of a line bundle. Trivial determinant forces that line bundle to be torsion, thereby converting projectively finite monodromy into finite linear monodromy.

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